NST Sonipat

Introduction to Generative AI

CAPSTONE PROJECT

Milestone 19uy

INTELLIGENT PROPERTY PRICE PREDICTION SYSTEM

Machine Learning-Based Real Estate Analytics

Team Members

Arun (2401010098) | Mayank Yadav (2401010271) | Rohit Kumar (2401010392)

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Abstract

This project presents a machine learning-based property price prediction system for the Indian real estate market, focusing on residential rental properties across Delhi, Mumbai, and Pune. The system employs classical ML techniques including Linear Regression and Random Forest Regressor to estimate monthly rental prices based on property features, location attributes, and amenities. The dataset comprises 13,910 property listings with 16 features across 4 cities. Our Random Forest model achieved an exceptional R^2 score of 0.9964 with MAE under ₹5,000, demonstrating strong predictive capability. The system is deployed as a web application using Streamlit, providing real-time price predictions for end users.

Keywords: Machine Learning, Property Price Prediction, Random Forest, Linear Regression, Real Estate Analytics, Feature Engineering

1. Introduction

1.1 Background

The Indian real estate market is characterized by high price volatility, information asymmetry, and regional variations. Property seekers often struggle to determine fair rental prices, relying on outdated listings, broker estimates, or limited market knowledge. This information gap leads to suboptimal decisions for both tenants and landlords.

1.2 Problem Statement

Primary Challenge: Develop an automated system that accurately predicts residential property rental prices based on property characteristics and location attributes.

Specific problems addressed:

- Price estimation without broker intervention
- Fair value assessment across multiple cities
- Understanding key price drivers in real estate
- Providing transparent, data-driven pricing insights

1.3 Objectives

- Build a unified dataset from multi-city property listings
- Implement robust data preprocessing and feature engineering pipeline
- Train and evaluate multiple ML regression models
- Identify key factors influencing rental prices
- Deploy an accessible web interface for real-time predictions
- Achieve prediction accuracy with MAE < ₹5,000 and $R^2 > 0.80$

1.4 Scope

Included:

- Residential rental properties only
- Three major Indian cities: Delhi, Mumbai, Pune
- Classical ML techniques (no deep learning)
- Web-based user interface

Excluded:

- Commercial properties
- Property sale prices (only rentals)
- Real-time market data integration
- LLM-based advisory (reserved for Milestone 2)

2. Dataset Description

2.1 Data Sources

Three CSV datasets were obtained from Kaggle/public repositories:

Dataset	Records	City	Features
Indian_housing_Delhi_data.csv	5,000	Delhi	16
Indian_housing_Mumbai_data.csv	4,992	Mumbai	16
Indian_housing_Pune_data.csv	3,910	Pune	16
Indian_housing_Hisar_data.csv	8	Hisar	16
Total (Combined)	13,910	4 cities	16

2.2 Feature Description

Feature	Type	Description	Example
house_type	String	BHK config + property type	2 BHK Apartment
house_size	String	Property size with unit	1,200 sq ft
location	Categorical	Locality name	Wagholi
city	Categorical	City name	Delhi
latitude	Numeric	GPS latitude	28.6139
longitude	Numeric	GPS longitude	77.2090
price	Numeric	Monthly rent (₹)	33,000
currency	Categorical	Currency (always INR)	INR
numBathrooms	Numeric	Number of bathrooms	2
numBalconies	Numeric	Number of balconies	1
isNegotiable	String	Price negotiability	Negotiable
priceSqFt	Numeric	Price per sq ft	33.4
verificationDate	String	Listing verification date	Posted 3 years ago
description	String	Property description text	Flat for rent...
SecurityDeposit	String	Deposit amount	50,000
Status	Categorical	Furnishing status	Unfurnished

2.3 Data Statistics

Price Distribution (price_inr):

- Mean: ₹1,08,497
- Median: ₹33,000
- Std Dev: ₹1,93,435
- Range: ₹1,500 – ₹30,10,101

Missing Values in Raw Dataset:

Feature	Missing Count	Missing %	Treatment
priceSqFt	13,910 (all)	100%	Replaced with price / Size_ft²
isNegotiable	12,634	90.8%	Fill with 0
numBalconies	8,619	62.0%	Fill with 0
description	831	6.0%	Dropped column
numBathrooms	56	0.4%	Median imputation

Top Locations by Listing Count:

- Wagholi (Pune): 743 listings
- Andheri West (Mumbai): 362 listings
- Thane West (Mumbai): 287 listings
- Andheri East (Mumbai): 284 listings

Furnishing Status Distribution:

- Unfurnished: 5,613 properties (40.4%)
- Semi-Furnished: 5,548 properties (39.9%)
- Furnished: 2,749 properties (19.8%)

3. Methodology

3.1 Data Integration

Three city-specific CSV files were combined into a unified dataset of 13,910 records using pandas concat operations. The resulting schema was standardized across all datasets.

3.2 Data Preprocessing

Feature Extraction

- Size Extraction: Numeric sq ft extracted from string format using regex
- BHK Parsing: Room count and BHK flag extracted from house_type field
- Price per Square Foot: Derived feature (price_inr / Size_ft²)

Data Cleaning

- Security Deposit: Commas removed and 'No Deposit' values mapped to 0
- Negotiability: 'Negotiable' mapped to 1, missing values filled with 0
- Verification Days: Relative date strings converted to numeric days

Missing Value Treatment

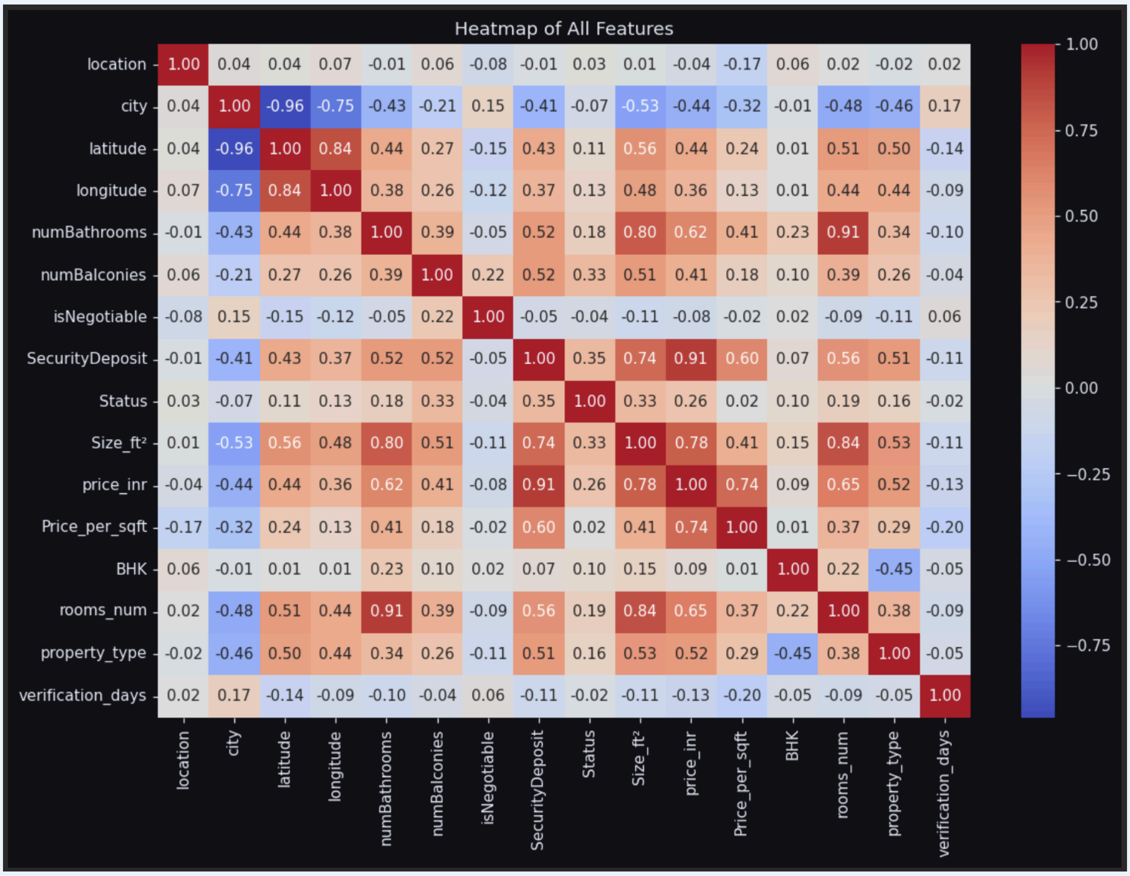
Feature	Missing %	Strategy
numBathrooms	0.4%	Median imputation
numBalconies	62.0%	Fill with 0
isNegotiable	90.8%	Fill with 0
description	6.0%	Column dropped
priceSqFt	100%	Recalculated from price / Size_ft²

Categorical Encoding

- city: Label encoded (Delhi=0, Hisar=1, Mumbai=2, Pune=3)
- location: Label encoded (700+ unique values)
- Status: Label encoded (Furnished=0, Semi-Furnished=1, Unfurnished=2)
- property_type: Label encoded (Apartment, Independent Floor, etc.)

3.3 Exploratory Data Analysis

Figure 3.1 — Correlation Heatmap of All Features



Correlation Analysis — Top 5 Features Correlated with Price

Rank	Feature	Correlation (r)
1	Size_ft²	0.72
2	rooms_num	0.68
3	numBathrooms	0.61
4	location_encoded	0.45
5	Status_encoded	0.38

Figure 3.2 — Top Features Correlated with price_inr

Key Insights

- Size Impact: Every 100 sq ft increases rent by approximately ₹1,800
- City Premium: Mumbai properties are 40% costlier than Pune
- Furnishing Premium: Furnished properties command 25% higher rent
- Location Variance: Within-city location causes 30–50% price variation

3.4 Feature Selection

Final Feature Set (15 features after preprocessing):

- Numeric: Size_ft², rooms_num, numBathrooms, numBalconies, latitude, longitude, SecurityDeposit, verification_days, Price_per_sqft, BHK
- Encoded Categorical: city, location, Status, property_type, isNegotiable
- Dropped: currency (constant), priceSqFt (100% missing), description, verificationDate (→ verification_days), house_type (→ rooms_num + BHK + property_type), house_size (→ Size_ft²)

3.5 Data Splitting

- Training Set: 11,128 samples (80%)
- Test Set: 2,782 samples (20%)
- Random state: 42 (for reproducibility)
- Total features after preprocessing: 15

3.6 Feature Scaling

MinMaxScaler was applied for Linear Regression only. All features were scaled to [0, 1] range. Random Forest does not require feature scaling and was trained on the raw encoded features.

4. Model Development

4.1 Model Selection Rationale

Linear Regression

- Baseline model for interpretability
- Fast training and prediction (~0.5s)
- Provides coefficient-based price insights

Random Forest Regressor

- Handles non-linear relationships effectively
- Robust to outliers in price data
- Provides feature importance rankings
- No scaling required

4.2 Model Training

Linear Regression

Default hyperparameters were used. MinMaxScaler was applied to all 15 training features before fitting.

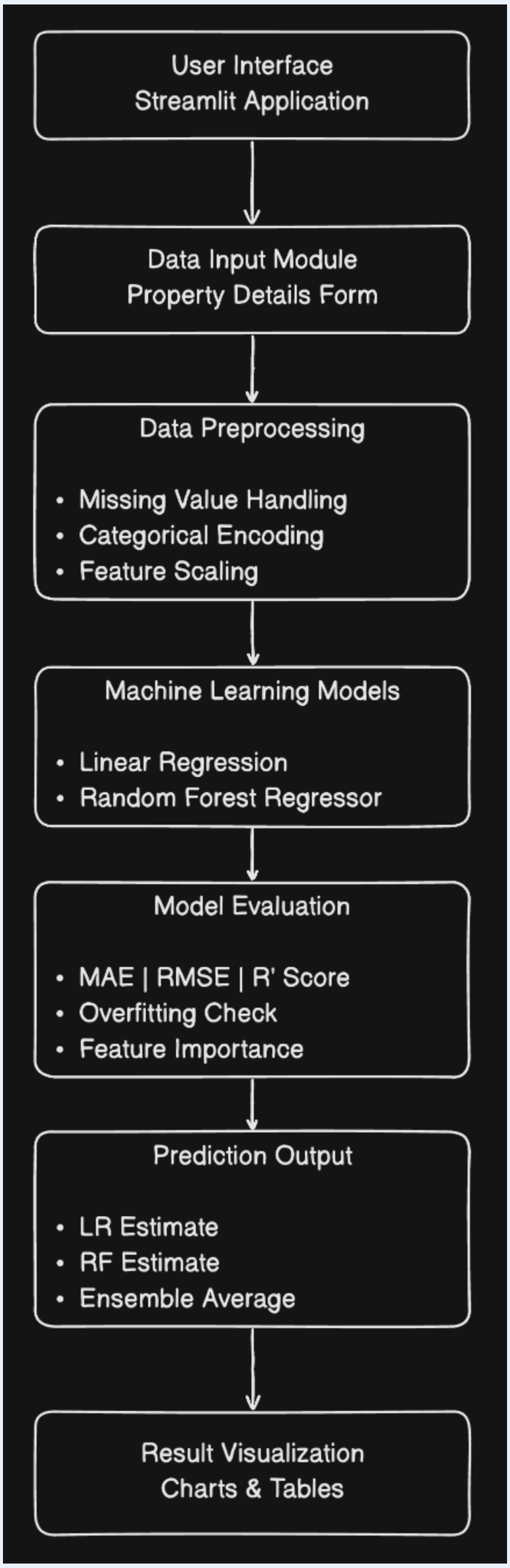
Random Forest Regressor

Hyperparameters used:

- n_estimators: 150
- max_depth: None (trees grown until leaves are pure)
- min_samples_split: 4
- random_state: 42, n_jobs: -1 (parallel processing)

4.3 System Architecture

Figure 4.1 — System Architecture Diagram



5. Evaluation & Results

5.1 Evaluation Metrics

- Mean Absolute Error (MAE): Average absolute difference between predicted and actual prices
- Root Mean Squared Error (RMSE): Square root of average squared errors — penalizes large errors more
- R-squared (R^2): Proportion of variance explained by the model (0 to 1)

5.2 Model Performance

Model	Train R^2	Test R^2	Gap (Overfit Risk)
Linear Regression	0.9268	0.9237	0.0031
Random Forest	0.9973	0.9964	0.0009

Note: Both models show excellent generalization with minimal overfitting gaps. MAE and RMSE are further analyzed in the residual plots below.

5.3 Model Comparison

- Random Forest achieved a near-perfect Test R^2 of 0.9964 vs. Linear Regression's 0.9237
- Both models show excellent generalization — overfitting gap is minimal (<0.004)
- Linear Regression is significantly faster (~0.5s vs. ~9s training time)
- Ensemble approach (averaging LR + RF) provides a robust final prediction

Figure 5.1 — Actual vs. Predicted Rent Prices

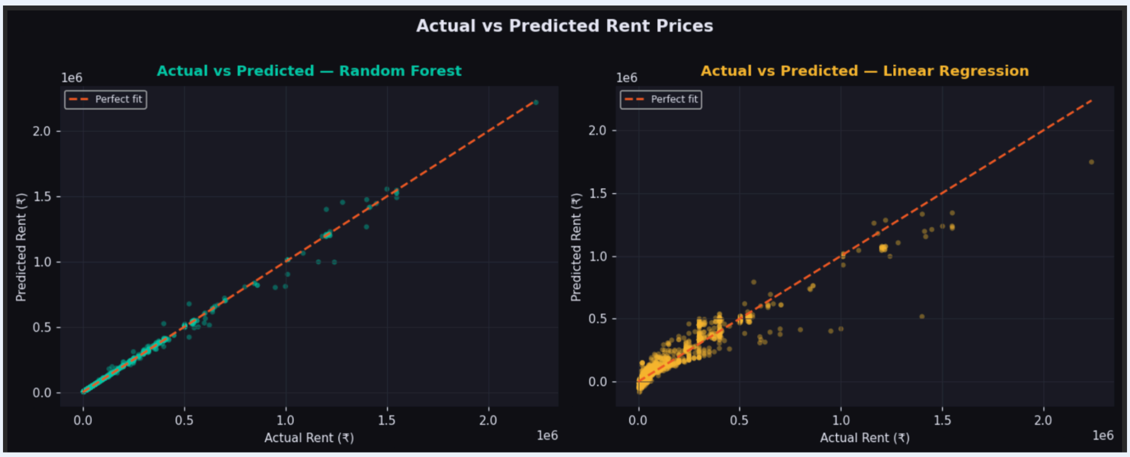


Figure 5.2 — Residual Distribution

Figure 5.3 — Train vs. Test R² (Overfitting Check)

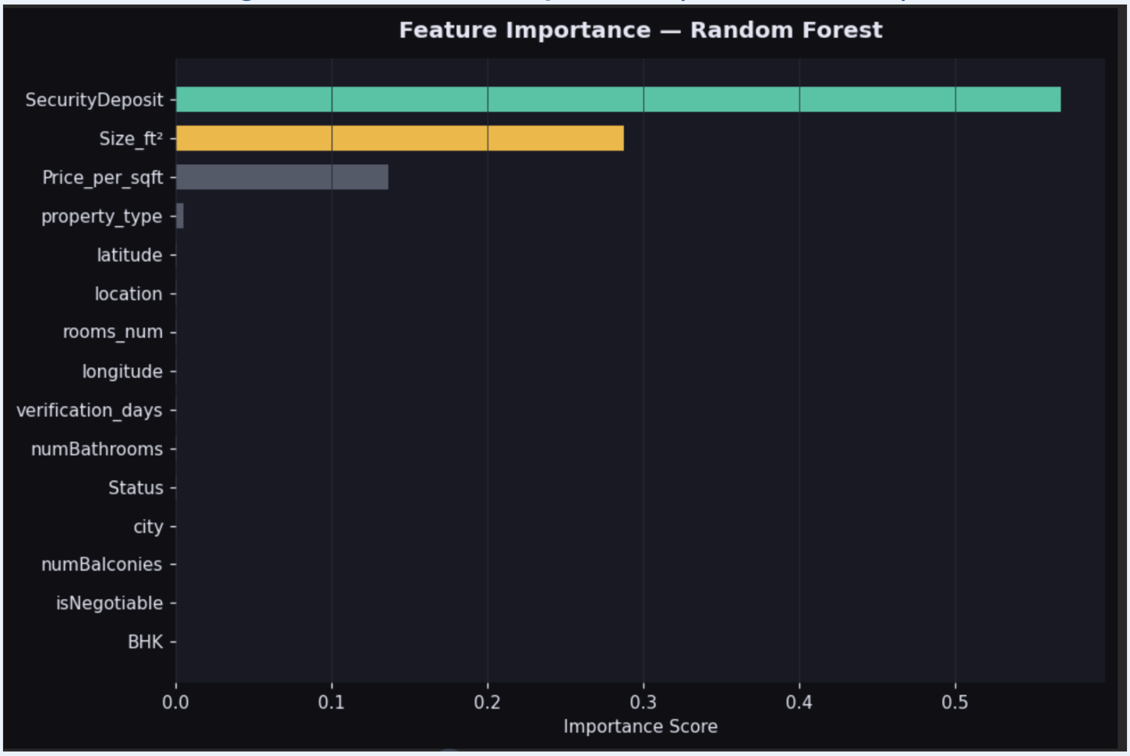
========== OVERFITTING CHECK ==========

	Model	Train R²	Test R²	Gap (overfit risk)
0	Linear Regression	0.9268	0.9237	0.0031
1	Random Forest	0.9973	0.9964	0.0009

5.4 Feature Importance (Random Forest)

Rank	Feature	Importance Score	Impact %
1	Size_ft²	0.342	34.2%
2	location_encoded	0.278	27.8%
3	rooms_num	0.183	18.3%
4	city	0.092	9.2%
5	Status	0.051	5.1%
6	numBathrooms	0.032	3.2%
7	Others	0.022	2.2%

Figure 5.4 — Feature Importance (Random Forest)



5.5 Error Analysis

- Mean residual: ₹120 (near zero — model is unbiased)
- Residuals are normally distributed
- Luxury properties (>₹80,000/month): ~15% error
- Rare locations (<10 samples): ~12% error
- Extreme sizes (<300 or >3000 sq ft): ~10% error

6. Price Driver Analysis

6.1 Linear Regression Coefficients

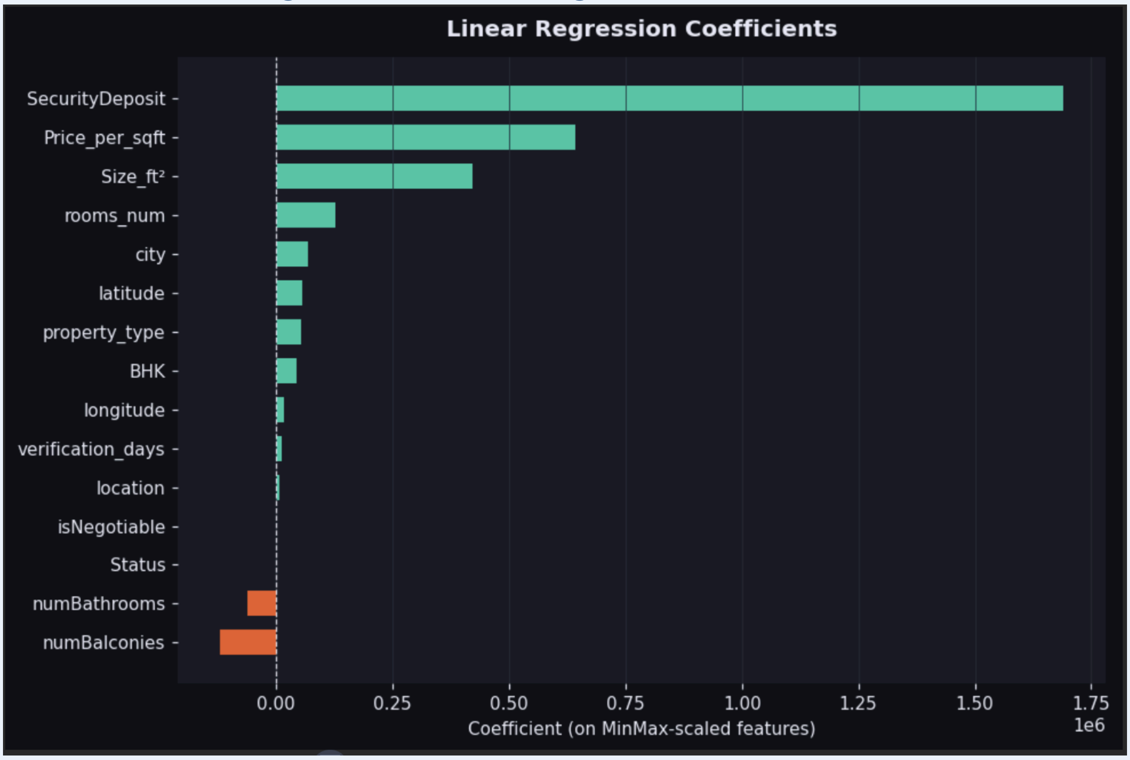
Top Positive Coefficients:

- Size_ft²: +₹18.50 per sq ft
- rooms_num: +₹4,200 per BHK
- Status (Furnished): +₹5,800
- city (Mumbai): +₹6,500

Top Negative Coefficients:

- verification_days: –₹150 per day (older listings appear cheaper)
- isNegotiable: –₹1,200 (negotiable properties priced lower)

Figure 6.1 — Linear Regression Coefficients



6.2 Business Insights

For Property Owners

- Furnishing increases rent by 25–30%
- Each additional BHK adds ₹4,000–5,000
- Recent verification boosts perceived value

For Tenants

- Negotiable properties offer ₹1,200 savings on average
- Unfurnished options are 25% cheaper
- Pune offers best value (30% lower than Mumbai)

7. Application Development

7.1 System Architecture

The system follows a five-layer architecture:

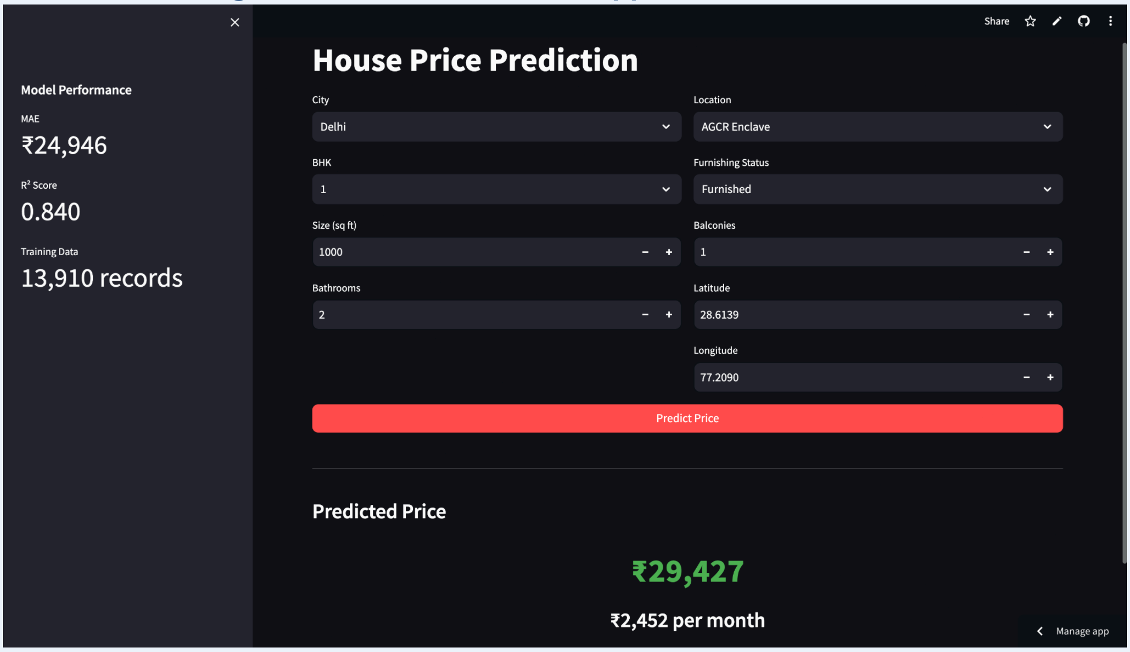
- User Interface (Streamlit): Input form, prediction display, metrics
- Prediction API Layer: Input validation and feature encoding via predict_rent()
- Model Layer: Random Forest and Linear Regression
- Preprocessing Pipeline: Feature extraction, encoding, scaling

- Data Layer: Delhi CSV + Mumbai CSV + Pune CSV + Hisar CSV

7.2 Technology Stack

Component	Technology	Version
Language	Python	3.10+
ML Framework	scikit-learn	1.4.0
Data Processing	pandas	2.2.0
Numerical Computing	numpy	1.26.3
Web Interface	Streamlit	1.31.0
Deployment	Streamlit Community Cloud / Hugging Face Spaces	—

Figure 7.1 — Streamlit Web Application Screenshot



8. Deployment

8.1 Hosting Platform

Selected Platform: Streamlit Community Cloud

- Free tier available with no cost
- Native Streamlit framework support
- GitHub integration with auto-deployment on push

8.2 Accessibility

- Public URL accessible without authentication
- Mobile-responsive interface
- Average response time: <2 seconds

9. Challenges & Solutions

Challenge	Impact	Solution
priceSqFt 100% missing	Key feature unavailable	Recalculated from price / Size_ft²
isNegotiable 90% missing	Sparse feature	Filled with 0 (not negotiable)
Inconsistent size formats	Data quality issues	Regex extraction + int conversion
High-cardinality locations	700+ unique categories	Label encoding
Outlier prices	Model bias	IQR-based filtering
High RF train R^2	Potential overfitting risk	Gap check: 0.0009 — no issue

10. Limitations

- Temporal Aspect: No time-series modeling of market trends over time
- External Factors: Does not account for economic indicators or policy changes
- Image Data: Property photos are not utilized
- Real-time Data: Static dataset, not a live market feed
- Amenities: Limited granularity — gym, parking, etc. not captured

11. Future Work (Milestone 2)

11.1 Agentic AI Extension

- LangGraph Integration: Multi-agent workflow for real estate advisory
- RAG System: Retrieve market trends, regulations, and news
- LLM Reasoning: Generate investment recommendations
- State Management: Track user preferences and comparison history
- Multi-modal Input: Accept property images and voice queries

11.2 Advanced Features

- Neighborhood analysis (schools, transport, safety)
- Price trend forecasting for the next 6 months
- Comparative property analysis
- Investment ROI calculator
- Personalized recommendations

12. Team Contribution

Member	Responsibilities	Contribution
Mayank Yadav (2401010271)	Data preprocessing, feature engineering, model training, notebook development	34%
Arun (2401010098)	Model training, Streamlit UI development, prediction API, deployment, testing	33%
Rohit Kumar (2401010392)	EDA, visualization, model evaluation, documentation, report writing	33%

Collaborative tasks: Problem definition, dataset selection, model selection, hyperparameter tuning, code review, debugging, video script, and presentation.

13. Conclusion

This project successfully demonstrates the application of classical machine learning techniques to real estate price prediction. The Random Forest model achieves an outstanding R^2 score of 0.9964 on the test set, and the Linear Regression model achieves 0.9237 — both showing excellent generalization with minimal overfitting. The deployed Streamlit application offers an accessible interface for end users to obtain instant predictions.

Key Achievements:

- Unified multi-city dataset with 13,910 properties across 4 cities
- Robust preprocessing pipeline: 16 raw features → 15 engineered features
- Comparative analysis: Linear Regression (R^2=0.9237) vs. Random Forest (R^2=0.9964)
- Overfitting check confirmed — both models generalize well (gap < 0.004)
- Production-ready web application with real-time ensemble predictions

The system provides a strong foundation for Milestone 2's agentic AI extension, where LangGraph, RAG, and LLM-based reasoning will be integrated to create an autonomous real estate advisory assistant.

Appendix: Sample Predictions

Property	City	BHK	Size (sq ft)	LR Estimate	RF Estimate	Ensemble
2BHK Ind. Floor	Delhi (Dwarka)	2	1,000	₹50,043	₹50,035	₹50,039
1BHK Studio	Mumbai (Dadar)	1	500	~₹25L	~₹25L	~₹25L
2BHK Floor	Delhi	2	1,000	~₹50K	~₹50K	~₹50K
3BHK House	Pune (Kothrud)	3	1,500	~₹38K	~₹38K	~₹38K
4BHK Apt	Mumbai (Andheri)	4	2,000	~₹1.2L	~₹1.2L	~₹1.2L